

YOLO Model Testing Report

Model:	yolov9m
Model Size:	39.1 MB
Parameters:	20.2 M
FLOPs (640x640):	77.87 GFLOPs
Run Name:	yolov9m_testing_20251124_094248
W&B Run Name:	yolov9m_bdd100k_yolo_test_20251124_094248
Timestamp:	2025-11-24 09:50:44
Dataset:	bdd100k_yolo - test split
Images Processed:	20000
Total Execution Time	439.72s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	17.07 ms per image
FPS (Frames Per Second)	58.59

Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.3105
Recall	0.6244
F1-Score	0.4148
mAP@0.5	0.0992

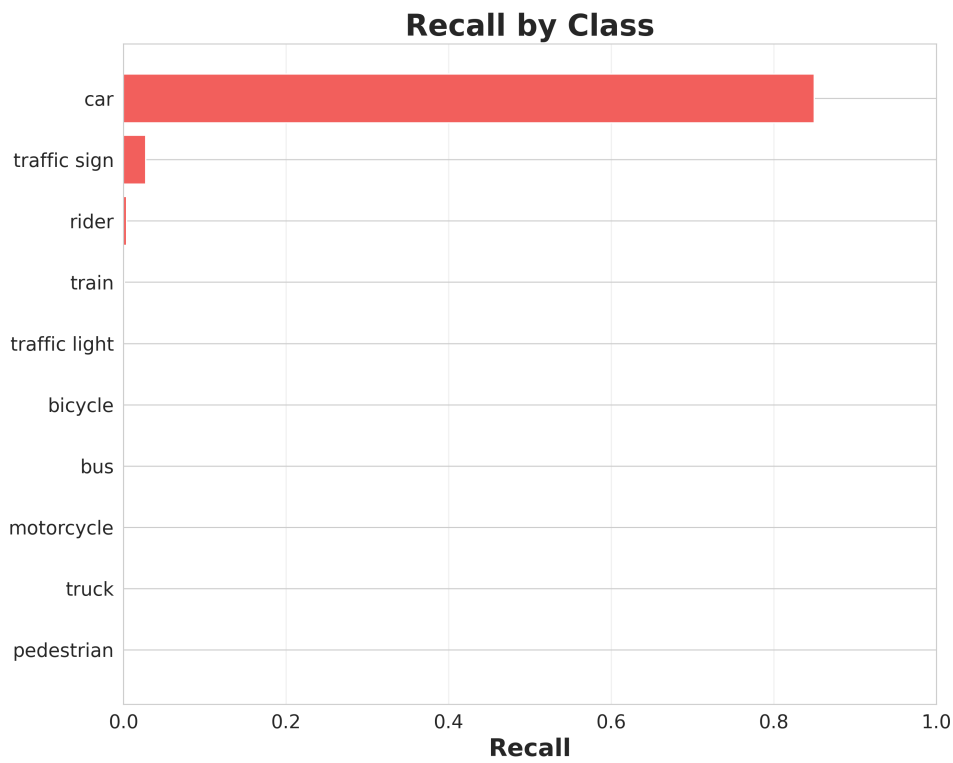
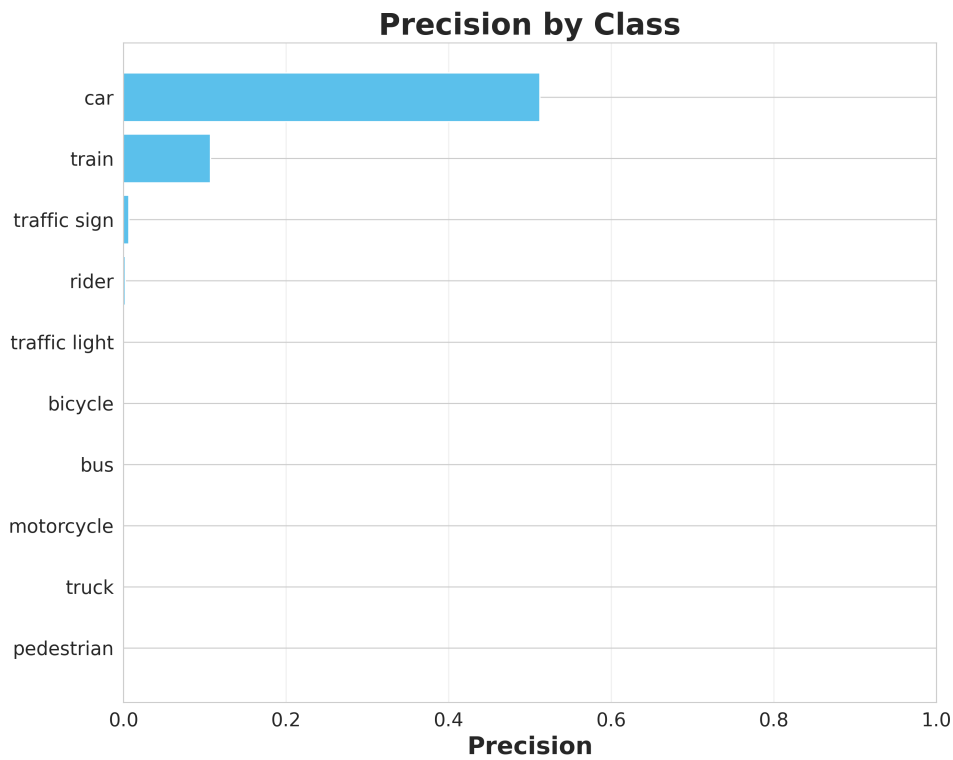
mAP@0.5:0.95	0.0582
True Positives	105614
False Positives	234527
False Negatives	63525

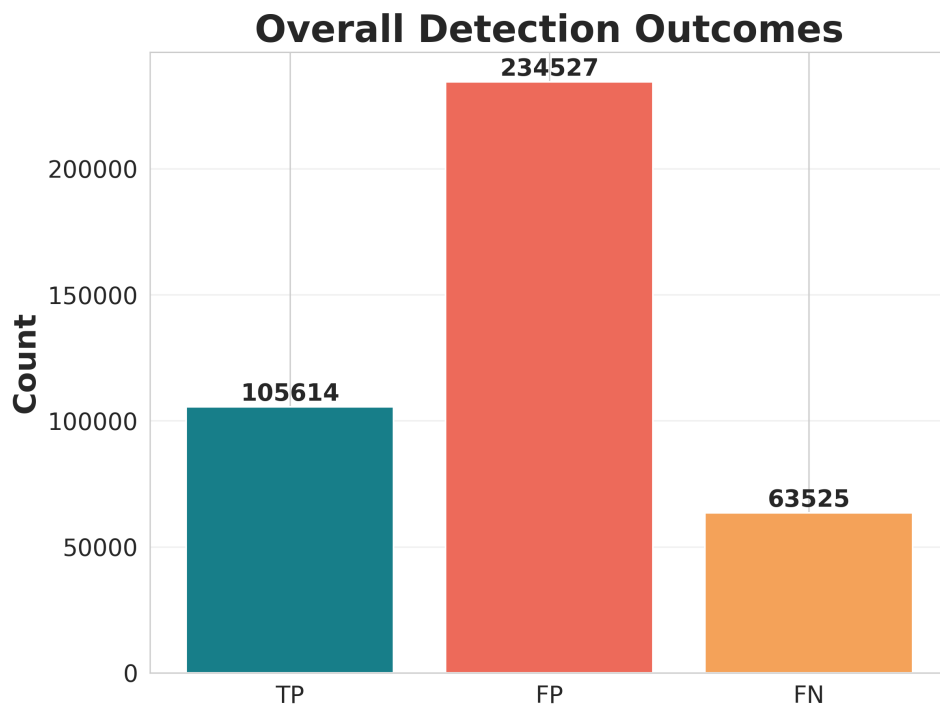
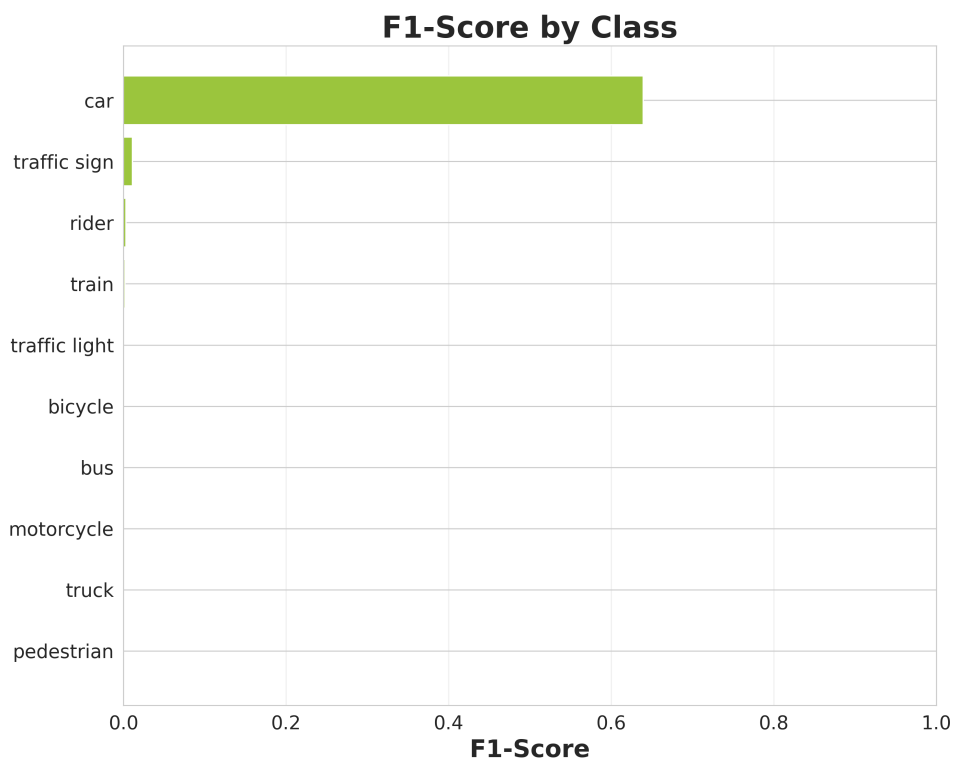
Confusion Matrix

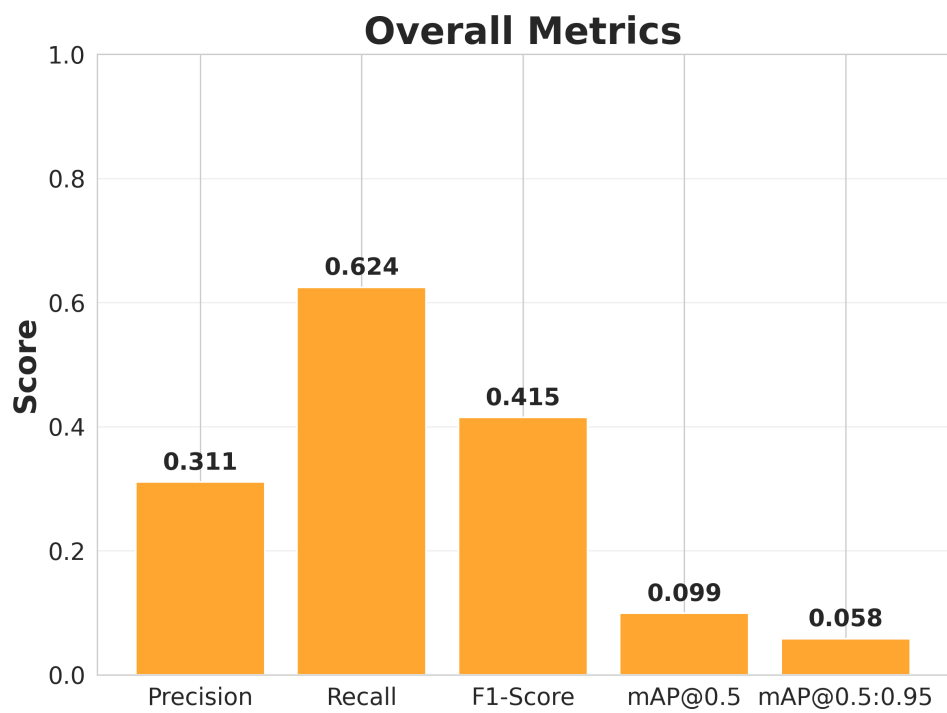
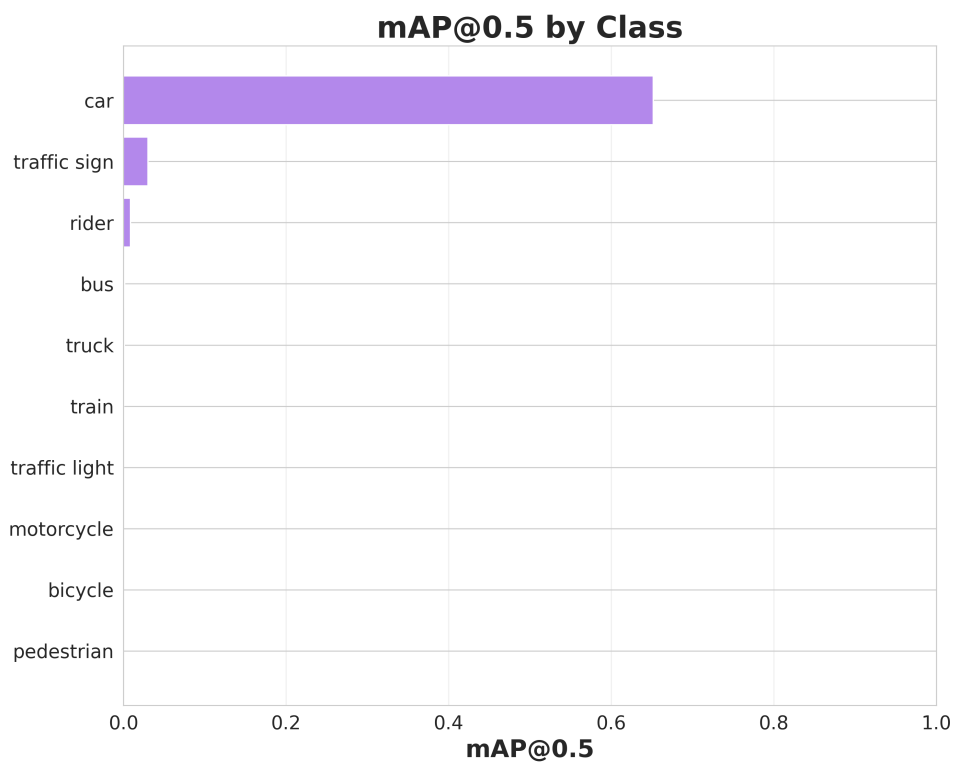
Confusion Matrix (yolov9m validation)											
True Class	pedestrian	655	79	1				3	13		
	rider	3	2								
	car	13	105134	976	226			20	67		
	truck	11	12								
	bus										
	train		469	440	1348	3			4		
	motorcycle		4	11	10	5					
	bicycle	1	3248	3787	350	2		2	21		
	traffic light		1	1							
	traffic sign		3	1		2		11272	474		
		pedestrian	rider	car	truck	bus	train	motorcycle	bicycle	traffic light	traffic sign
Predicted Class											

Correct Predictions (Diagonal Sum): 105614
Total Matched Predictions: 382415

Performance Visualizations







Per-Class Performance

Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
pedestrian	0	0	13395	0.0000	0.0000	0.0000	0.0000
rider	3	1290	721	0.0023	0.0041	0.0030	0.0087
car	105134	99959	18546	0.5126	0.8500	0.6396	0.6519
truck	0	8701	435	0.0000	0.0000	0.0000	0.0013
bus	0	3217	18	0.0000	0.0000	0.0000	0.0017
train	3	25	3015	0.1071	0.0010	0.0020	0.0003
motorcycle	0	0	147	0.0000	0.0000	0.0000	0.0000
bicycle	0	0	10416	0.0000	0.0000	0.0000	0.0000
traffic light	0	52817	12	0.0000	0.0000	0.0000	0.0001
traffic sign	474	68518	16820	0.0069	0.0274	0.0110	0.0302

Sample Predictions: Ground Truth vs Model

Sample #1: d1e75f9e-c0479527.jpg

Ground Truth Objects: 13 | Predicted Objects: 4

Weather: partly cloudy | Scene: highway | Time of Day: daytime



Sample #2: e81dca70-b6927573.jpg

Ground Truth Objects: 7 | Predicted Objects: 6

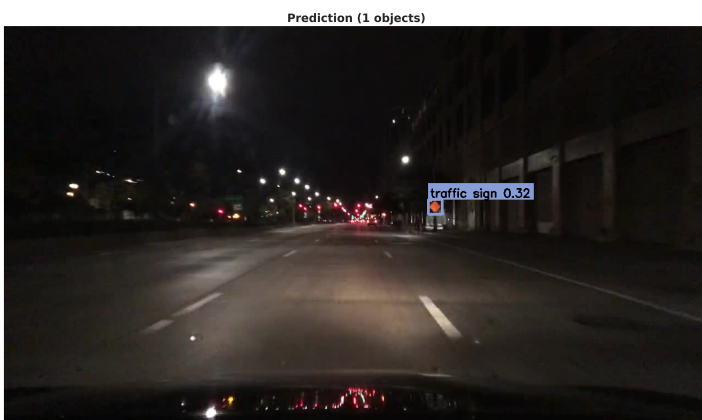
Weather: rainy | Scene: highway | Time of Day: night



Sample #3: f0de5a46-d00c9dee.jpg

Ground Truth Objects: 12 | Predicted Objects: 1

Weather: clear | Scene: city street | Time of Day: night



Sample #4: e526b869-cf93ace2.jpg

Ground Truth Objects: 10 | Predicted Objects: 5

Weather: clear | Scene: highway | Time of Day: night

Ground Truth (10 objects)



Prediction (5 objects)



Sample #5: de78c347-9e3c6b3c.jpg

Ground Truth Objects: 15 | Predicted Objects: 9

Weather: snowy | Scene: city street | Time of Day: night

Ground Truth (15 objects)



Prediction (9 objects)



Sample #6: cbbfa4d0-9c3f6e67.jpg

Ground Truth Objects: 14 | Predicted Objects: 10

Weather: clear | Scene: highway | Time of Day: daytime

Ground Truth (14 objects)



Prediction (10 objects)



Generated by YOLO Quick Test Script